

Strategic Product Placement Analysis

1. INTRODUCTION

❖ 1.1 Project Overview

Strategic Product Placement Analysis is a data visualization project aimed at understanding the impact of product positioning on retail sales performance. The project analyzes factors such as product category, promotions, competitor pricing, seasonal trends, and customer foot traffic to identify optimal product placement strategies. Using Tableau Desktop and Tableau Public, interactive dashboards and stories were created to provide business insights and support data-driven decision-making.

❖ 1.2 Purpose

The purpose of this project is to help retailers improve product visibility, increase customer engagement, and maximize sales revenue by identifying effective product placement strategies using data visualization techniques.

2. IDEATION PHASE

❖ 2.1 Problem Statement

Retail businesses often face difficulties in determining the best placement strategy for products within stores. Improper product placement can lead to low visibility, poor customer engagement, and reduced sales performance. This project aims to analyze product placement factors and provide recommendations for improving retail sales.

❖ 2.2 Empathy Map Canvas

The empathy map was developed by considering the perspective of retail store managers and business owners.

Think and Feel

- Improve sales performance.
- Increase customer satisfaction.
- Optimize product placement.

Hear

- Customer feedback.
- Market trends.
- Sales team suggestions.

See

- Competitor strategies.
- Seasonal demand changes.
- Product sales variations.

Say and Do

- Conduct promotions.
- Rearrange products.
- Monitor sales reports.

Pain

- Low sales.
- Poor product visibility.
- Inefficient shelf utilization.

Gain

- Increased revenue.
- Better customer experience.
- Data-driven decisions.

2.3 Brainstorming

Several project ideas were considered including customer behavior analysis, sales forecasting, promotion effectiveness analysis, inventory management, and product placement optimization. After evaluation, Strategic Product Placement Analysis was selected because of its business relevance and visualization potential.

3. REQUIREMENT ANALYSIS

❖ 3.1 Customer Journey Map

1. Identify sales-related problems.
2. Collect product sales data.
3. Analyze customer purchasing behavior.
4. Create visualizations.
5. Generate insights.
6. Implement recommendations.
7. Monitor performance improvements.

❖ 3.2 Solution Requirement

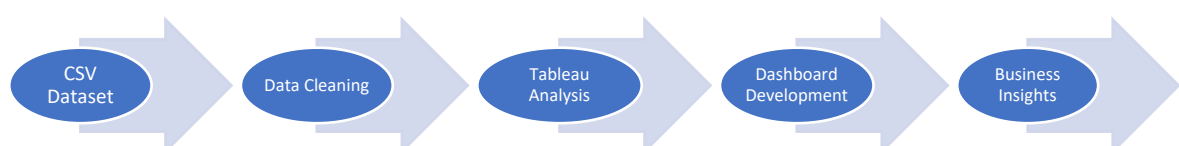
Functional Requirements

- Import CSV dataset.
- Analyze sales data.
- Create visualizations.
- Build dashboards.
- Create story pages.
- Publish results.

Non-Functional Requirements

- Usability
- Reliability
- Performance
- Availability
- Scalability
- Maintainability

❖ 3.3 Data Flow Diagram



❖ 3.4 Technology Stack

Component	Technology
Dataset	CSV
Visualization Tool	Tableau Desktop
Publishing Platform	Tableau Public
Repository	GitHub
Documentation	PDF

4. PROJECT DESIGN

❖ 4.1 Problem Solution Fit

Customer Segment

- Retail Store Managers
- Business Analysts
- Marketing Teams

Customer Constraints

- Large volume of sales data.
- Difficulty identifying trends manually.

Available Solutions

- Excel reports.
- Manual analysis.

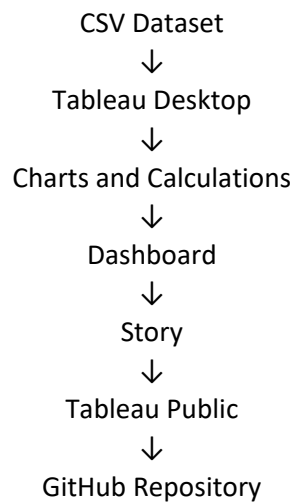
Proposed Solution

Interactive Tableau dashboards for product placement analysis.

❖ 4.2 Proposed Solution

The project uses Tableau dashboards and stories to analyze product positioning, promotions, competitor pricing, and seasonal trends. The generated insights support effective decision-making and improve sales performance.

❖ 4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

❖ 5.1 Project Planning

Week	Activity
Week 1	Topic selection and planning
Week 2	Dataset collection
Week 3	Data cleaning and preparation
Week 4	Visualization development
Week 5	Dashboard creation
Week 6	Story development
Week 7	Testing and publishing
Week 8	Documentation and submission

6. FUNCTIONAL AND PERFORMANCE TESTING

❖ 6.1 Performance Testing

➤ Accuracy Testing

Verified dashboard results with source dataset values.

➤ Performance Testing

Measured dashboard loading speed and responsiveness.

➤ Usability Testing

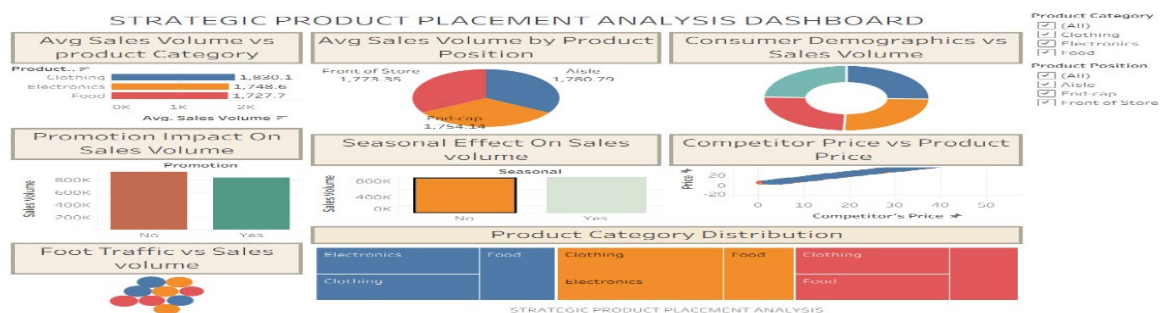
Ensured dashboard navigation and interaction were simple and intuitive.

➤ Compatibility Testing

Verified functionality across Tableau Desktop, Tableau Public, and web browsers.

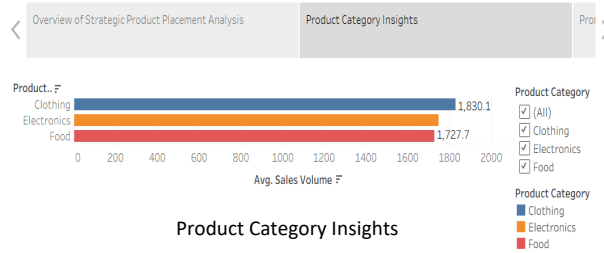
7. RESULTS

❖ 7.1 Output Screenshots

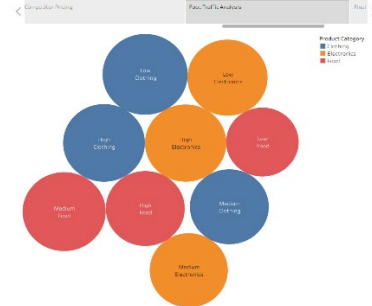


Dashboard Image

Story 1



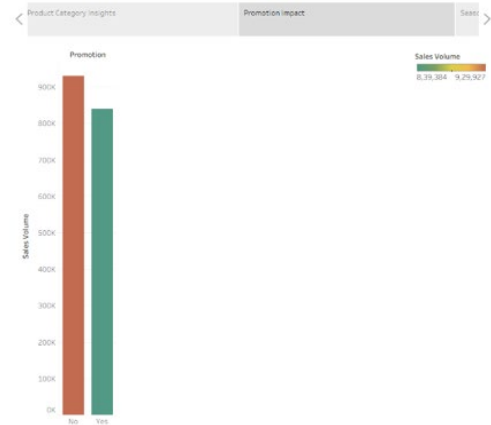
Story 1



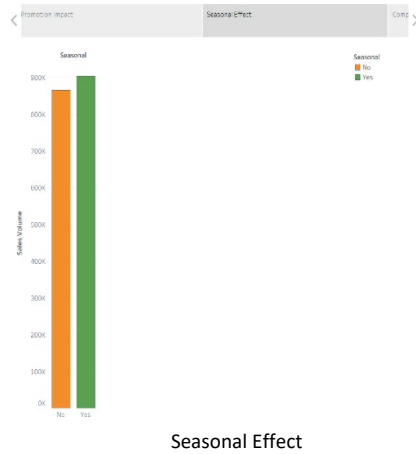
Story 1



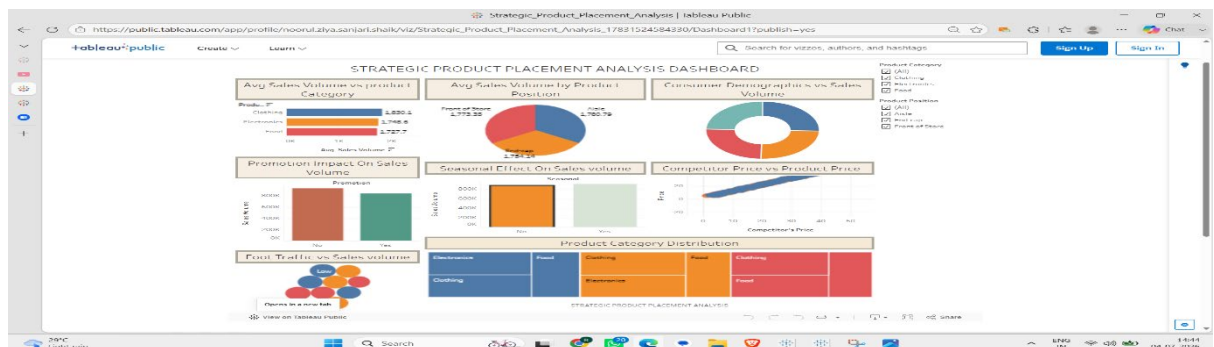
Story 1



Story 1



Promotion Impact



Tabelu_Public_Web

8. ADVANTAGES & DISADVANTAGES

❖ Advantages

- Improves product visibility.
- Supports data-driven decisions.
- Enhances customer experience.
- Increases sales opportunities.

❖ Disadvantages

- Depends on historical data quality.
- Requires periodic updates.
- Results may vary between stores.

9. CONCLUSION

The Strategic Product Placement Analysis project successfully demonstrates how data visualization can help retailers optimize product placement strategies and improve sales performance. Tableau dashboards and stories provided meaningful insights that support business decision-making.

10. FUTURE SCOPE

- Integration with real-time sales systems.
- AI-based recommendation systems.
- Predictive sales forecasting.
- Multi-store comparative analysis.
- Advanced customer segmentation.

11. APPENDIX

❖ Dataset Link

- Dataset link: <https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales>

❖ GitHub & Project Demo Link

- GitHub Repository Link: <https://github.com/sanjari4013/Strategic-Product-Placement-Analysis>
- Tableau Public Link: <https://public.tableau.com/app/profile/noorul.ziya.sanjari.shaik>
- https://public.tableau.com/app/profile/noorul.ziya.sanjari.shaik/viz/Strategic_Product_Placement_Analysis_17831524584330/Dashboard1?publish=yes